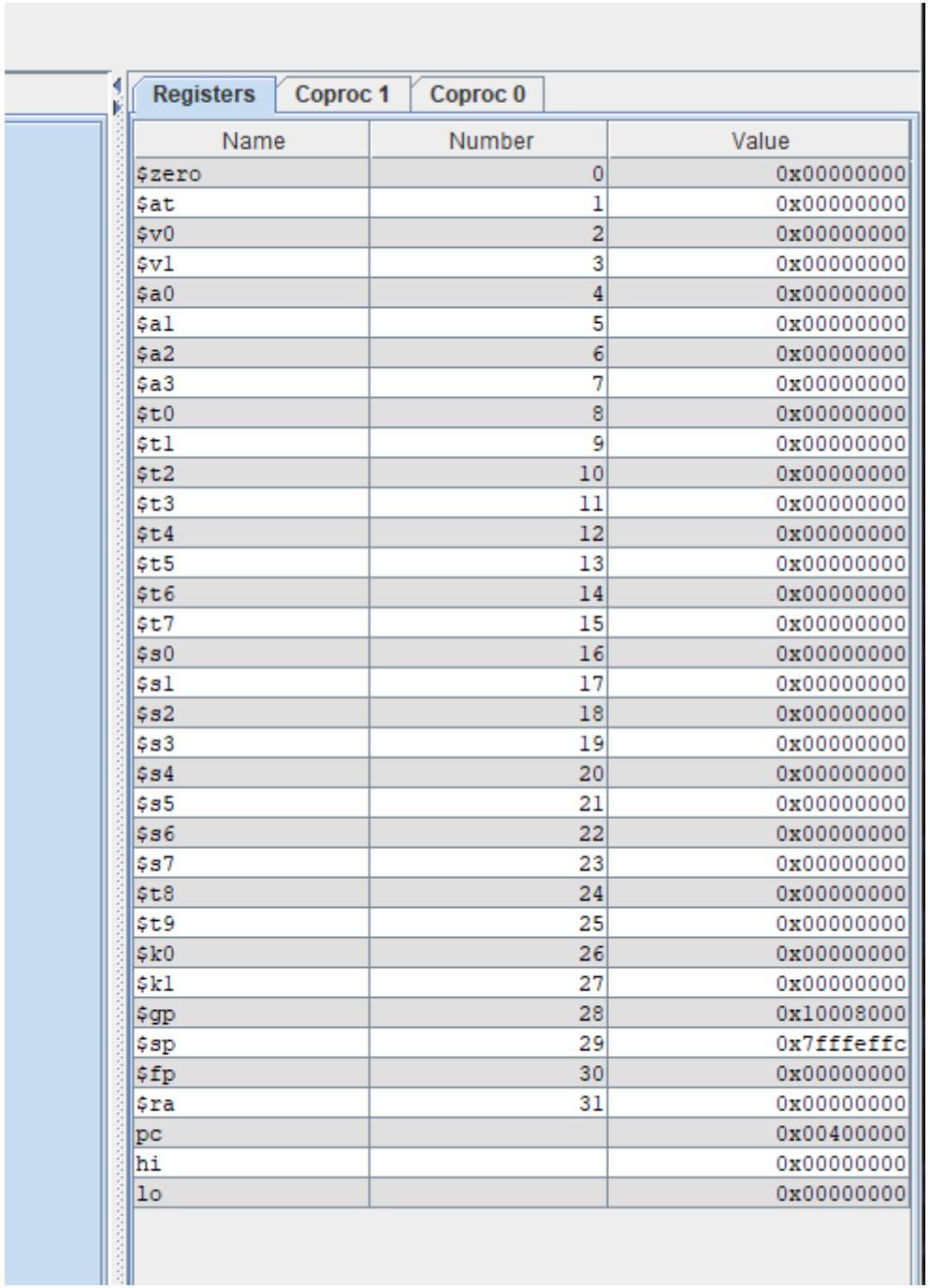


Registers | Coproc 1 | Coproc 0

Name	Number	Value
$zero	0	0x00000000
$at	1	0x00000000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x00000000
$t2	10	0x00000000
$t3	11	0x00000000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400000
hi		0x00000000
lo		0x00000000

COM203 - Computer Architecture

Lab #1

Mars



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Mars Lab 1

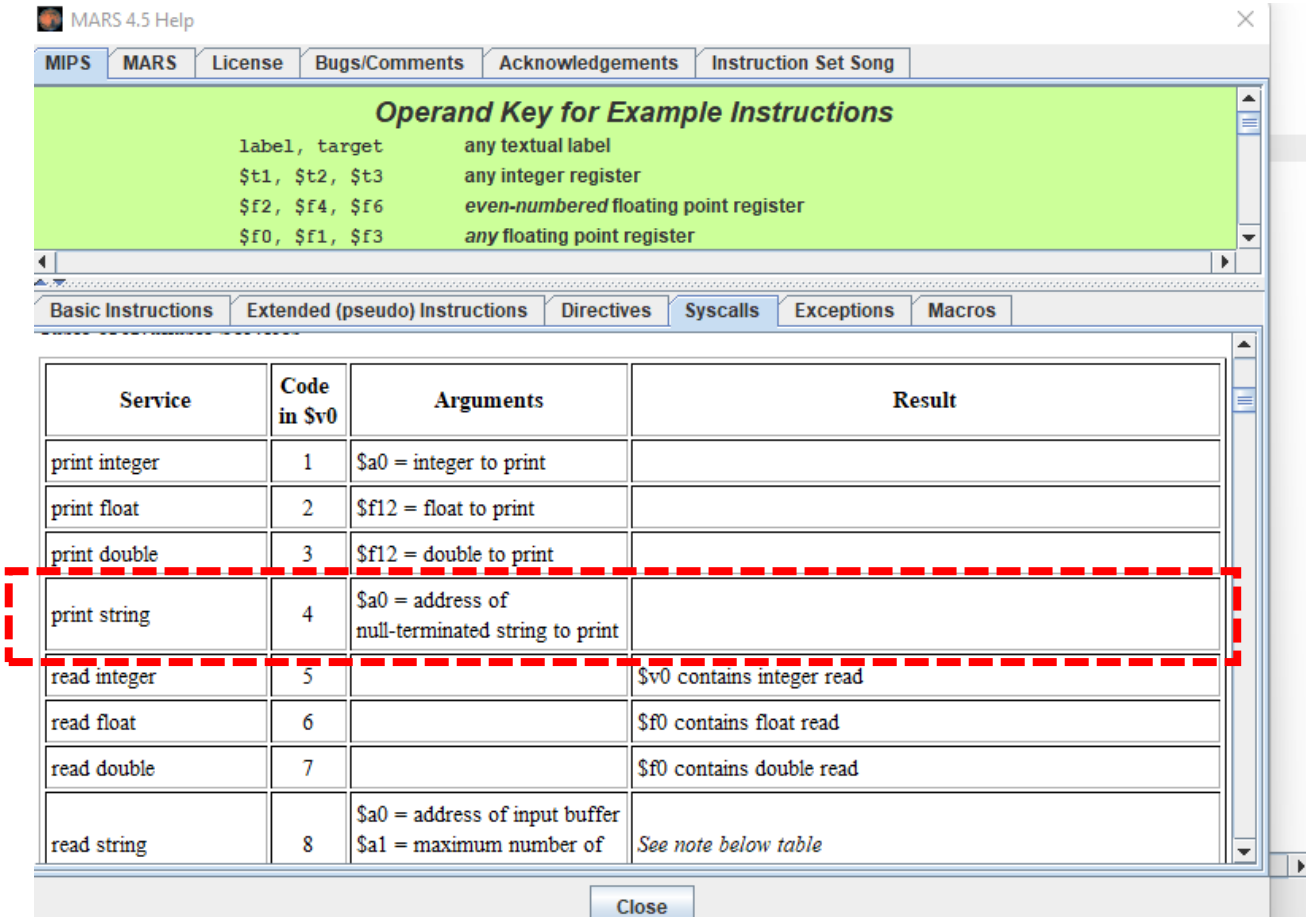
Write a program in assembly language that can print ”Hello world”

Program:

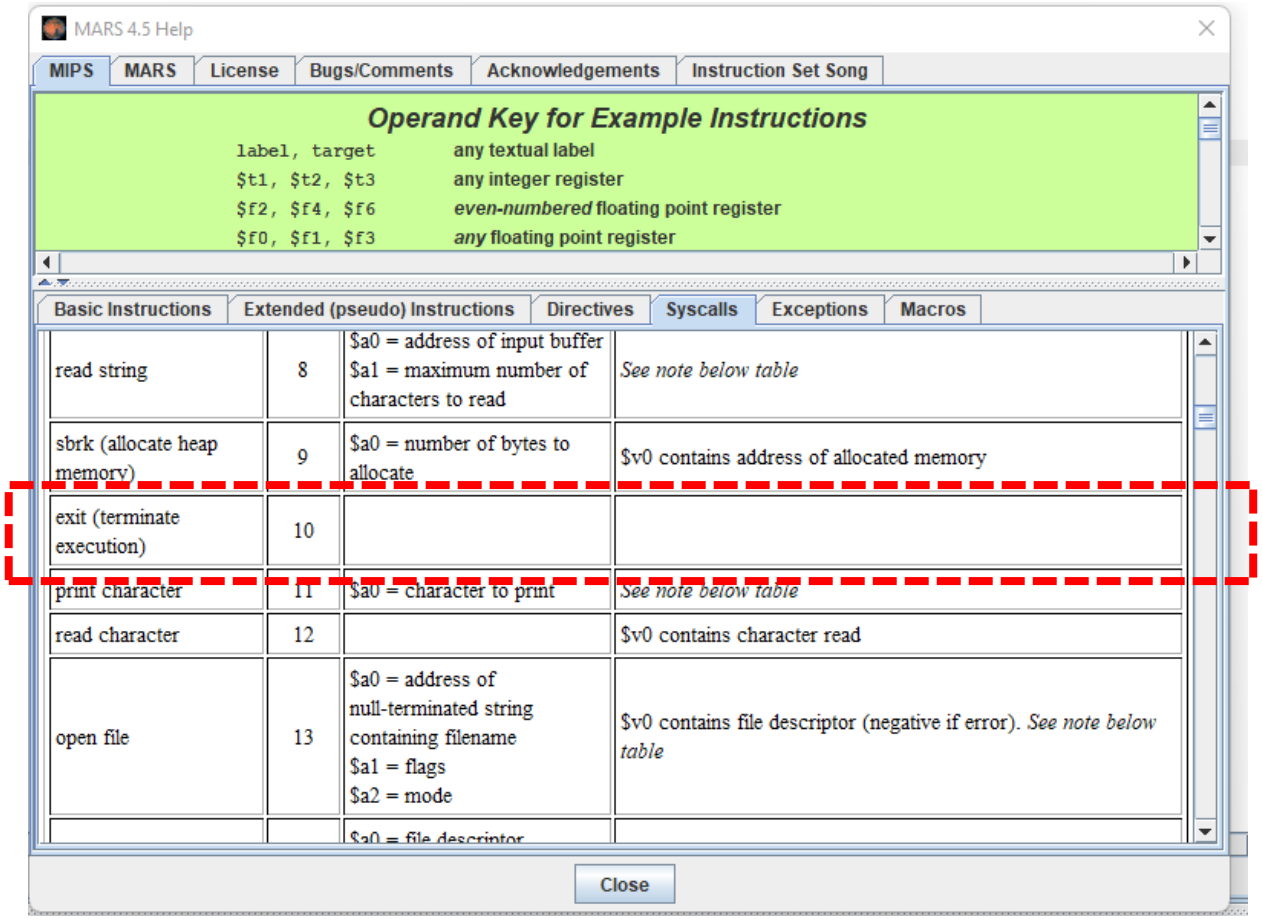
```
.data
        myText: .asciiz "Hello world \n"

.text
        li $v0, 4
        la $a0, myText
        Syscall
        li $v0, 10
        syscall
```

Print String



Service	Code in $v0	Arguments	Result
print integer	1	$a0 = integer to print	
print float	2	$f12 = float to print	
print double	3	$f12 = double to print	
print string	4	$a0 = address of null-terminated string to print	
read integer	5		$v0 contains integer read
read float	6		$f0 contains float read
read double	7		$f0 contains double read
read string	8	$a0 = address of input buffer $a1 = maximum number of	*See note below table*


MARS 4.5 Help
MIPS
MARS
License
Bugs/Comments
Acknowledgements
Instruction Set Song
Operand Key for Example Instructions
label, target any textual label
$t1, $t2, $t3 any integer register
$f2, $f4, $f6 even-numbered floating point register
$f0, $f1, $f3 any floating point register
Basic Instructions
Extended (pseudo) Instructions
Directives
Syscalls
Exceptions
Macros
read string 8 $a0 = address of input buffer $a1 = maximum number of characters to read See note below table
sbrk (allocate heap memory) 9 $a0 = number of bytes to allocate $v0 contains address of allocated memory
exit (terminate execution) 10
print character 11 $a0 = character to print See note below table
read character 12 $v0 contains character read
open file 13 $a0 = address of null-terminated string containing filename $a1 = flags $a2 = mode $v0 contains file descriptor (negative if error). See note below table
$a0 = file descriptor
Close